Using Google Colab

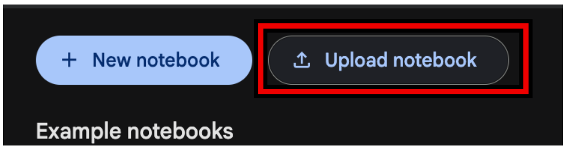
The course now provides scripts that run in Google Colab. You need a Google account and a web browser to acces it.

If you have issues with GitHub Codespace, we recommend to switch over. Everything you learned for codespace/jupyter notebook still applies. What changed is where they run, and how you give them your data-provider logins. This handout covers both aspects.

1. Open a notebook

Colab opens notebooks straight from GitHub. You do not download anything and you do not need a GitHub account.

1. Go to https://colab.research.google.com.
2. Click **Upload notebook**. Despite the name, this is what opens the dialog you need — you are not uploading anything.

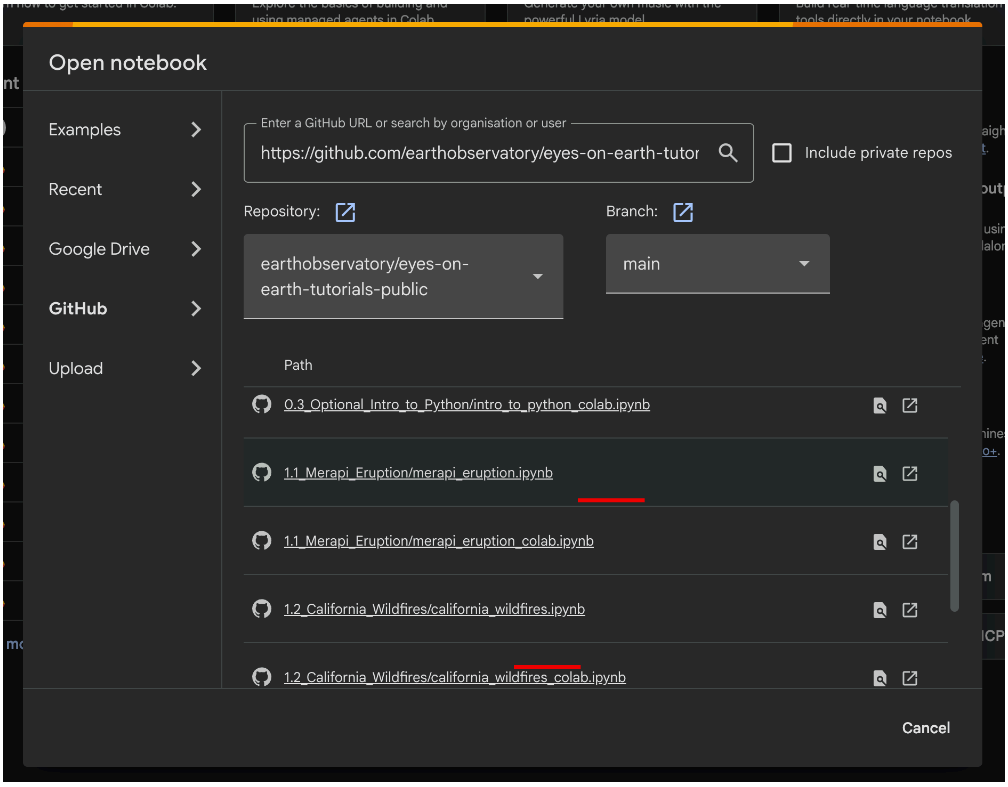


Click Upload notebook. If you already have a notebook open, File → Open notebook does the same thing.

3. The **Open notebook** popup appears. Choose **GitHub** in the left-hand list.
4. Paste this into the search box and press Enter:

```
https://github.com/earthobservatory/eyes-on-earth-tutorials-public
```

5. Set the rest as in the picture below: **Repository** on earthobservatory/eyes-on-earth-tutorials-public, **Branch** on `main`, and **Include private repos** left unticked.
6. Click the notebook you want to open. **Always take the one whose name ends in _colab.**



The Open notebook dialog, with GitHub chosen on the left. Paste the URL at the top, check Repository and Branch, then click a notebook in the list.

That last point is the one to remember. Every tutorial exists twice:

File	Use it?
`california_wildfires_colab.ipynb`	**Yes** — set up for Colab
`california_wildfires.ipynb`	No — expects an environment Colab does not have

They contain the same tutorial. The _colab one has an extra cell at the top that installs what Colab is missing and reads your logins, and its wording matches what you actually see on screen. The plain one is for a pre-built environment, and its first data cell will fail here.

> **Use the GitHub tab, not the Upload tab.** They sit next to each other in that dialog. The GitHub tab reads the notebook out of the repository at the moment you open it. The Upload tab puts a fixed copy in your Drive, which never changes again.

Your own copy

The notebook you open from GitHub is **read-only**. The moment you change anything, use **File → Save a copy in Drive**. That copy is yours, it lives in your Google Drive, and it is the one to keep working in — Colab will otherwise lose your edits when you close the tab.

> **Nothing syncs with GitHub in either direction.** Opening from GitHub gives you the notebook as it was at that moment; it is not a live link. Your edits stay in your Drive copy and never go back to the repository, and a tab you left open yesterday will not pick up a correction made today.

So, if you are told a tutorial has been fixed, do not reload your Drive copy, it keeps the old version. Open it again from the GitHub picker, which is where the fix will be.

Anything you have already written into your own copy stays there. Save the new version separately with **File → Save a copy in Drive**.

2. Work through them in this order

	Notebook	What it is
1	`0.2.1_Account_Check_colab.ipynb`	**Do this first.** Sets up your logins and checks all three work.
2	`intro_to_python_colab.ipynb`	Optional. Skip it if you are comfortable with Python.
3	`merapi_eruption_colab.ipynb`	Sentinel-2 imagery of the 2023 Merapi eruption.
4	`download_from_copernicus_colab.ipynb`	Downloads a Sentinel-2 scene from Copernicus.
5	`california_wildfires_colab.ipynb`	Eaton Fire NDVI and NBR.

> **The list will grow as the course goes on.** Later tutorials are added to the repository as we reach them, so open the GitHub picker again each time rather than working from a notebook you opened weeks ago — a tutorial that was not in the list last week may be there now.

There is no 0.1 for Colab. That notebook checks and repairs a conda environment, which Colab does not use. Each Colab notebook installs what it needs in its own first cell instead.

3. Your logins, once, in Colab Secrets

Three data providers need an account. Register for all three early,**the JAXA one is approved by a person and can take several working days.**

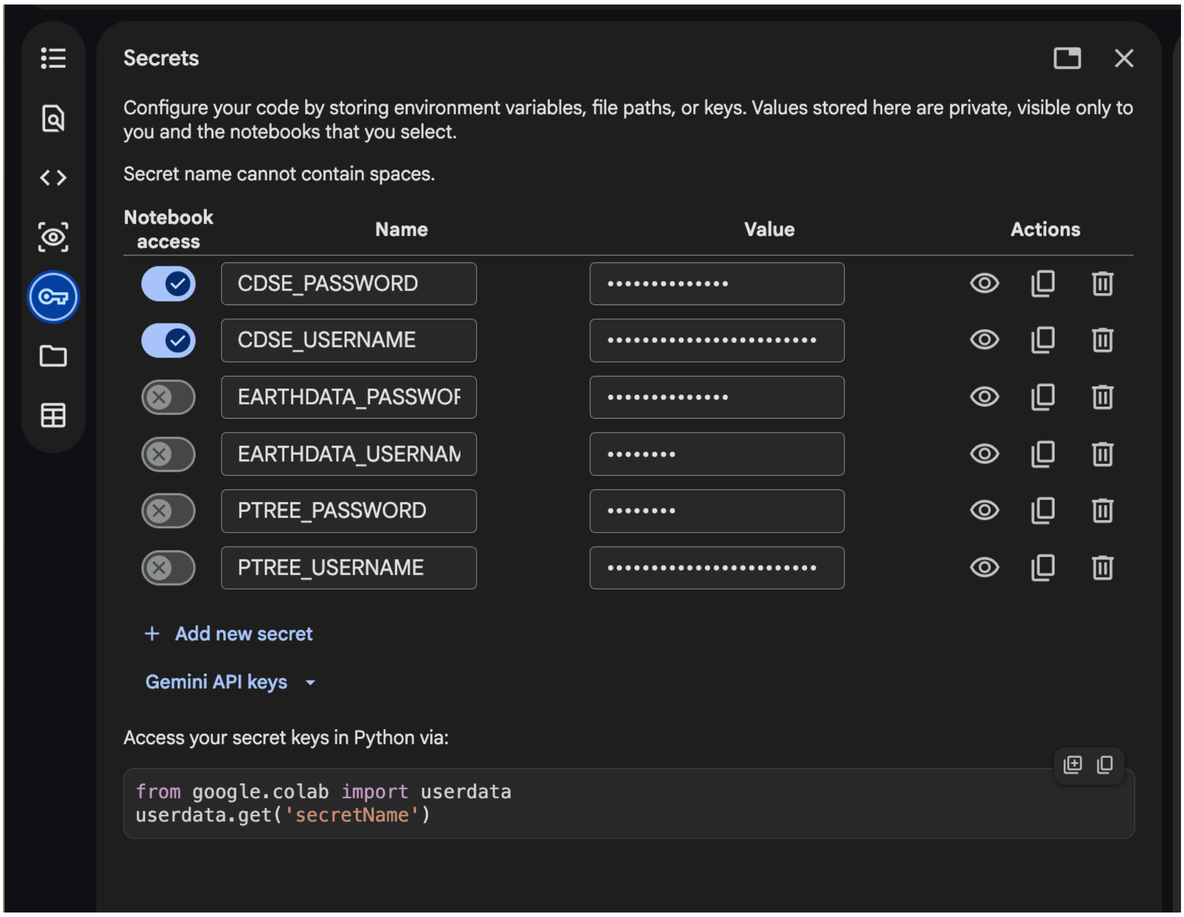
Provider	Register at	Approval
NASA Earthdata	urs.earthdata.nasa.gov/users/new	Immediate
JAXA P-Tree	eorc.jaxa.jp/ptree	**Several working days**
Copernicus Data Space	dataspace.copernicus.eu	Immediate

Colab throws its machine away when your session ends, so there is nowhere on it to keep a password. Instead, you put them in **Colab Secrets**, which belongs to your Google account and **outlives any session**.

Adding them

Open any _colab notebook, then:

7. Click the **key icon** in the left sidebar. The **Secrets** panel opens.
8. Click **+ Add new secret**.
9. Put the name in **Name**, exactly as written below. No spaces.
10. Paste the value in **Value** (which is your username or password).
11. Switch **Notebook access** on for that row.



The Secrets panel. Note the Notebook access switches: CDSE is on for this notebook, EARTHDATA and PTREE are off — those two would fail until the switch is turned on.

There are six:

Secret name	Value
`EARTHDATA_USERNAME`	Your NASA Earthdata username
`EARTHDATA_PASSWORD`	Your NASA Earthdata password
`PTREE_USERNAME`	Your JAXA P-Tree **FTP** username — usually your email with @ replaced by _
`PTREE_PASSWORD`	Your JAXA P-Tree **FTP** password
`CDSE_USERNAME`	The email address you registered with at Copernicus Data Space
`CDSE_PASSWORD`	Your Copernicus Data Space password

> **P-Tree's FTP credentials are not your P-Tree website login.** They arrive in the approval email. This is the single most common thing to get wrong.

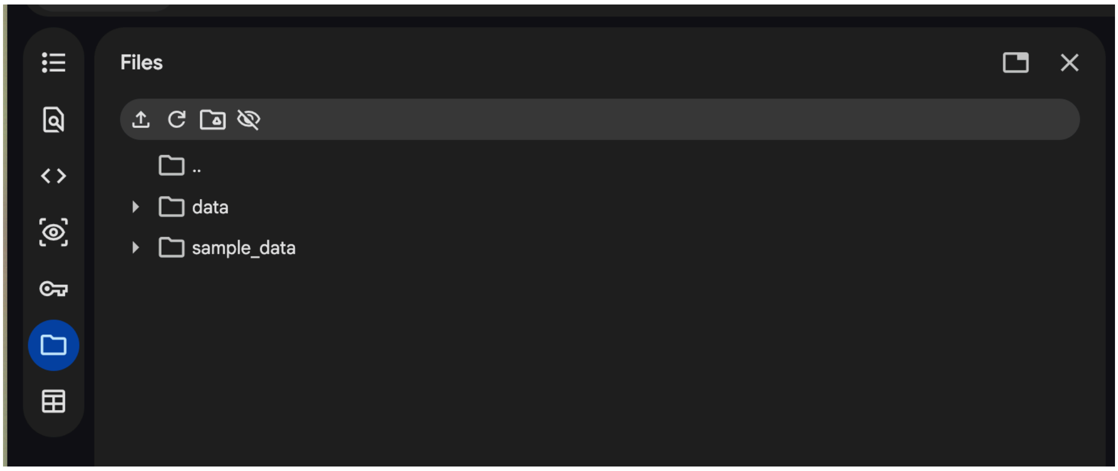
You add the secret **once**. But **Notebook access is per notebook**. This means that the first time you open a new tutorial, the switch will be off for the secrets, and the notebook will stop and tell you to grant access. Go back to the key icon and turn it on, or grant access when the code asks for it.

The account-check notebook (`0.2.1_Account_Check_colab.ipynb`) reports which secrets it can read. Please run that code first to check whether you have properly set-up the credentials

4. Everything you make disappears — save it

When your Colab session ends, the machine and every file on it are deleted. Downloaded data, figures, GeoTIFFs, all of it.

Download what you need for your write-up before you close the tab. Click the **folder icon** in the left sidebar, find the file, and use its three-dot menu → **Download**.



The Files panel, reached with the folder icon. Open a folder, then use a file's three-dot menu to download it.

Downloaded data is not worth saving as the notebooks can fetch it again in a few minutes. Your figures and GeoTIFFs are the ones you should save as it takes time to process, and they are the ones your assignment needs. So be mindful to always save the output when you are done with the notebook.

A session also ends if you leave it idle too long, so do not start a tutorial you have no time to finish.

5. If something goes wrong

- **A cell says a package will not import.** Do Runtime → Restart session, then run the setup cell again.
- **A secret is "not readable".** It is not added yet, or its Notebook access switch is off for this notebook. Both are fixed at the key icon.
- **The notebook cannot find your data.** Check you ran the setup cell — it creates the folders — and that your session has not restarted since you downloaded (which means you might need ot rerun your cells).
- **530 Login incorrect from JAXA.** You used your website login rather than the FTP credentials, or your registration is not approved yet.
- Anything else: see the S01 Troubleshooting page in the course repository.